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ORIGINAL RESEARCH

Population Health Model Predicting the Long-Term Impact of Sotatercept on Morbidity and Mortality in Patients with Pulmonary Arterial Hypertension (PAH)

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ABSTRACT

Introduction: Pulmonary arterial hypertension (PAH) is a rare, progressive disease associated with significant morbidity and mortality. The phase 3 STELLAR trial tested sotatercept plus background therapy (BGT) versus placebo plus BGT. BGT was comprised of mono-, double-, or triple-PAH targeted therapy. Building on STELLAR findings, we employed a population health model to assess the potential long-term clinical impact of sotatercept.

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Methods: Based on the well-established ESC/ERS 4-strata risk assessment approach, we developed a six-state Markov-type model (low risk, intermediate-low risk, intermediate-high risk, high risk, lung/heart-lung transplant, and death) to compare the clinical outcomes of sotatercept plus BGT versus BGT alone over a lifetime horizon. State-transition probabilities were obtained from STELLAR. Risk stratum-adjusted mortality and lung/heart-lung transplant probabilities were based on COMPERA PAH registry data, and the post-transplant mortality probability was obtained from existing literature. Model outcomes were discounted at 3% annually. Sensitivity analyses were conducted to examine model robustness.

Results: In the base case, sotatercept plus BGT was associated with longer life expectancy from model baseline (16.5 vs 5.1 years) versus BGT alone, leading to 11.5 years gained per patient. Compared with BGT alone, sotatercept plus BGT was further associated with a gain in infused prostacyclin-free life years per patient, along with 683 PAH hospitalizations and 4 lung/heart-lung transplant avoided per 1000 patients.

Conclusions: According to this model, adding sotatercept to BGT increased life expectancy by roughly threefold among patients with PAH while reducing utilization of infused prostacyclin, PAH hospitalizations, and lung/heart-lung transplants. Real-world data are needed to confirm these findings.

Trial Registration: ClinicalTrials.gov identifier, NCT04576988 (STELLAR).

Keywords: Pulmonary arterial hypertension; Risk strata; Sotatercept; STELLAR

Key Summary Points

Why carry out this study?

Pulmonary arterial hypertension (PAH) is a debilitating condition marked by its progressive nature with a significant burden on morbidity and mortality.

The phase 3 STELLAR trial has demonstrated the strong clinical benefit of sotatercept as an add-on treatment to stable background therapy for PAH compared to background therapy alone.

To enhance clinicians' understanding of the potential long-term treatment effectiveness, we developed a population health model based on STELLAR data to predict the long-term morbidity and mortality outcomes of sotatercept plus background therapy vs background therapy alone for patients with PAH.

What was learned from this study?

This study's findings reveal that adding sotatercept to stable background therapy has the potential to increase life expectancy by roughly threefold among patients with PAH, while reducing utilization of infused prostacyclin, PAH hospitalizations, and lung/heart-lung transplants.

INTRODUCTION

Pulmonary arterial hypertension (PAH) is a rare and progressive disease affecting an estimated 40,000 people in the US and 50,000 across Europe [1–3]. Patients living with PAH face considerable morbidity and mortality and experience poor prognoses as a result of increased pulmonary resistance, which leads to right ventricular failure and eventually death [4, 5]. Current median survival is 5–7 years after diagnosis [6–8].

Current treatment options for PAH include vasodilators tackling three distinct pathways (prostacyclin, nitric oxide, endothelin) [9]. Sotatercept, an activin signaling inhibitor, can address imbalances in activin/growth differentiation factors and the bone morphogenetic protein signaling. This novel activin receptor type IIA-Fc fusion protein traps activin 2a to improve cardiopulmonary function in patients with PAH and has demonstrated reverse pulmonary arterial wall and right ventricular remodeling [10, 11]. The phase 3 randomized controlled trial STELLAR (NCT04576988) demonstrated the clinical benefit of sotatercept as an add-on treatment to stable background therapy for PAH by showing an increase in the 6-min walking distance (6MWD) result by 40.8 m and time to clinical worsening reduced by 84% (hazard ratio [HR] 0.16, 95% CI 0.08–0.35). Background therapy was defined as the standard of care therapy for PAH at the time of conduct of STELLAR, including mono, double, or triple therapy of some combination of endothelin receptor antagonist (ERA), phosphodiesterase type-5 (PDE-5) inhibitor, soluble guanylate cyclase (sGC) stimulator, prostaglandin I2 (PGI2) analog, and PGI2 agonist [12].

The management of PAH involves lifelong therapeutic intervention, yet clinical evidence for pharmacotherapies in PAH is generally derived from clinical trials of short duration. In this context, assessments based on population health models can serve as a tool in aiding clinical decision-making [13]. By simulating patient disease progression over time, these models can facilitate the extrapolation of short-term trial data over longer periods.

Therefore, we developed a population health model aiming to evaluate the long-term health outcomes of sotatercept plus background therapy versus background therapy alone for the treatment of patients with PAH.

METHODS

Overview

We analyzed STELLAR, a phase 3 study of sotatercept, to derive patients' distribution and transition probabilities across risk strata at baseline and weeks 3, 12, and 24. Subsequently, data were derived from the COMPERA registry to understand behavior of patients with the respectively derived risk profile over time in a real-world cohort and to assess mortality and transplant [8]. A population health model was developed to simulate the disease course of PAH and compare long-term health outcomes associated with sotatercept plus background therapy vs background therapy alone in adult patients with symptomatic PAH (World Health Organization [WHO] functional class [FC] II or III), in line with the definition of the intent-to-treat (ITT) population and treatment arms of STELLAR (sotatercept plus background therapy vs background therapy alone).

The model adopted a lifetime horizon, assumed to be 30 years based on simulation results. These results revealed that a majority of patients, with a baseline mean age of 47.6 years, experienced mortality within this time frame across both arms. The model employed cycle lengths of 3 weeks (first cycle), 9 weeks (second cycle), and 12 weeks (third and subsequent cycles), in line with the visit schedule in STELLAR. Model outcomes were discounted at 3% annually beyond the first model cycle [14].

This study relied on previously conducted studies and did not involve any new studies requiring direct human subject participation; thus, ethics board approval was not required.

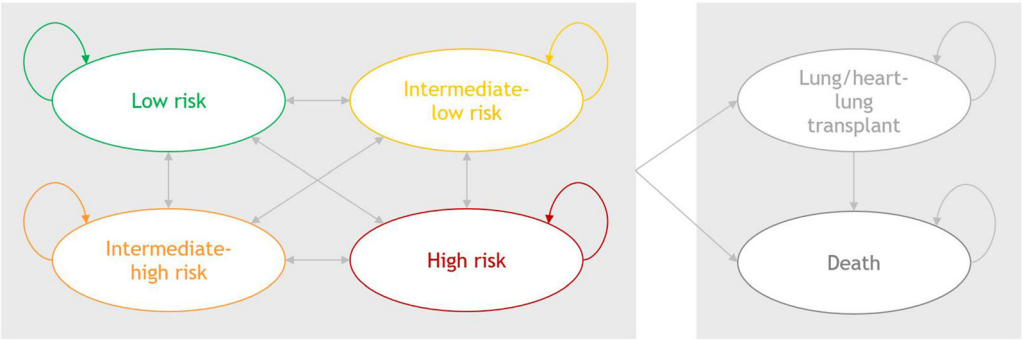

Fig. 1 Model structure

Model Structure

A Markov-type model was developed to simulate the disease course of PAH, comprising six mutually exclusive health states: low risk, intermediate-low risk, intermediate-high risk, high risk, lung/heart-lung transplant, and death (Fig. 1). These states were defined in line with the refined four-strata risk assessment, which has been validated across multiple PAH registries and is recommended in the 2022 European Society of Cardiology (ESC) and European Respiratory Society (ERS) treatment guideline for PAH because of its established prognostic relevance for survival [15]. Specifically, patients are categorized into one of the four risk strata, with those in the higher-risk group having a heightened risk of mortality, PAH hospitalization, and lung transplant [15].

At baseline, all patients enter the model via one of the four risk strata health states (Table 1 and Table 2). The cohort of patients was then followed through the model and could remain in their current health state, transition between the four risk strata health states, or progress to lung/heart-lung transplant or death during each model cycle, according to a set of transition probabilities obtained from STELLAR (Table 1). Across each model cycle, the proportion of patients occupying each health state accrued associated life-years (LYs) over time.

Model Parameters

Population Characteristics

The baseline patient characteristics of the modeled population are shown in Table 2. Additional patient characteristics that were not directly used as model inputs but used for validation purposes are shown in Supplementary Table 2.

Risk-strata Transition Probabilities

STELLAR data were used to calculate transition probabilities across risk strata (either worsening, improvement, or maintenance). Short-term transition probabilities used in the model (i.e., model baseline to week 3, week 3–12, and week 12–24) were based on observed patient transition counts across weeks 0, 3, 12, and 24 in STELLAR. Long-term transition probabilities (i.e., post week 24) were based on observed transition counts from weeks 12–24 in STELLAR, assuming that the short-term outcomes were generalizable to long-term outcomes.

Mortality, by Risk Stratum

Patients with PAH face an increased risk of mortality, with increasingly higher risk as the disease progresses, i.e., vascular remodeling advances [16, 17]. Therefore, to inform long-term all-cause mortality probabilities by risk stratum, parametric regression models were fitted to long-term 5-year survival curves from the COMPERA registry and the French PAH registry

Table 1 Distribution of risk as measured in STELLAR

Treatment group	From baseline	n^b	To Week 3			
			Low risk	Intermediate-low risk	Intermediate-high risk	High risk
Placebo (N^a = 160)	Low risk	43	40 (93.0%)	3 (7.0%)	0 (0.0%)	0 (0.0%)
	Intermediate-low risk	60	5 (8.3%)	48 (80.0%)	7 (11.7%)	0 (0.0%)
	Intermediate-high risk	38	0 (0.0%)	5 (13.2%)	33 (86.8%)	0 (0.0%)
	High risk	2	0 (0.0%)	0 (0.0%)	1 (50.0%)	1 (50.0%)
Sotatercept (N^a = 163)	Low risk	46	43 (93.5%)	3 (6.5%)	0 (0.0%)	0 (0.0%)
	Intermediate-low risk	60	13 (21.7%)	46 (76.7%)	1 (1.7%)	0 (0.0%)
	Intermediate-high risk	38	0 (0.0%)	17 (44.7%)	20 (52.6%)	1 (2.6%)
	High risk	0	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Treatment group	**From Week 3**	**n^b**	**To Week 12**			
			Low risk	**Intermediate Low Risk**	**Intermediate-high risk**	**High risk**
Placebo (N^a = 160)	Low risk	40	31 (77.5%)	9 (22.5%)	0 (0.0%)	0 (0.0%)
	Intermediate-low risk	54	7 (13.0%)	43 (79.6%)	4 (7.4%)	0 (0.0%)
	Intermediate-high risk	36	0 (0.0%)	7 (19.4%)	26 (72.2%)	3 (8.3%)
	High risk	1	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (100.0%)
Sotatercept (N^a = 163)	Low risk	54	49 (90.7%)	5 (9.3%)	0 (0.0%)	0 (0.0%)
	Intermediate-low risk	52	19 (36.5%)	31 (59.6%)	2 (3.8%)	0 (0.0%)
	Intermediate-high risk	20	2 (10.0%)	7 (35.0%)	11 (55.0%)	0 (0.0%)
	High risk	1	0 (0.0%)	1 (100.0%)	0 (0.0%)	0 (0.0%)
Treatment group	**From Week 12**	**n^b**	**To Week 24**			
			Low risk	**Intermediate low risk**	**Intermediate high risk**	**High risk**
Placebo (N^a = 160)	Low risk	40	35 (87.5%)	4 (10.0%)	1 (2.5%)	0 (0.0%)
	Intermediate-low risk	60	7 (11.7%)	43 (71.7%)	9 (15.0%)	1 (1.7%)

Table 1 continued

Treatment group	From Week 12	n[b]	To Week 24			
			Low risk	Intermediate low risk	Intermediate high risk	High risk
	Intermediate-high risk	25	0 (0.0%)	3 (12.0%)	18 (72.0%)	4 (16.0%)
	High risk	3	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (100.0%)
Sotatercept (N^a = 163)	Low risk	73	69 (94.5%)	4 (5.5%)	0 (0.0%)	0 (0.0%)
	Intermediate-low risk	48	14 (29.2%)	30 (62.5%)	4 (8.3%)	0 (0.0%)
	Intermediate-high risk	11	0 (0.0%)	6 (54.5%)	5 (45.5%)	0 (0.0%)
	High risk	0	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)

[a]Number of participants: full analysis set population

[b]Number of participants: full analysis set population, participants with non-missing risk assessment stratum at both analysis visits. Row percentages are based on this number

Source: Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR

Database date: 06 DEC 2022

*In the base case analysis, observed STELLAR data from week 12 to week 24 were used to inform the transition probabilities beyond 24 weeks

**Due to the small number of patients progressing to high risk in STELLAR, the data of sotatercept plus background therapy and background therapy alone were pooled together and then applied to both treatment arms, assuming that sotatercept has no treatment benefit on disease improvement in patients with high risk. This is a conservative assumption (i.e., in favor of the background therapy alone arm)

Table 2 Model input parameters

Parameter	Value	Distribution (SE) for probabilistic analysis	Source
Baseline patient characteristics			
Mean age, years	47.9	Not varied	STELLAR [12]
Female, %	79.3	Not varied	STELLAR [12]
Baseline patient distribution			
Low risk	29.6%	Dirichlet (2.55%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR
Intermediate-low risk	44.2%	Dirichlet (2.77%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR
Intermediate-high risk	25.2%	Dirichlet (2.42%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR
High risk	0.9%	Dirichlet (0.54%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR
Probabilities of transitioning across risk strata	See Supplementary Table 2	Dirichlet	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR
Probabilities of all-cause mortality by risk stratum	See Supplementary Figure 2	Multivariate normality for regression models	Rosenkranz et al. (2023) [17]
HR of all-cause mortality, sotatercept plus background therapy vs background therapy alone	0.25	Log-normal (0.81)*	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR
Probability of lung/heart-lung transplant, per 12-week cycle			
Low risk	0.0%	Not varied	Assumption**
Intermediate-low risk	0.0%	Not varied	Assumption**
Intermediate-high risk	0.1%	Beta (0.01%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). COMPERA
High risk	0.1%	Beta (0.01%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). COMPERA

Table 2 continued

Parameter	Value	Distribution (SE) for probabilistic analysis	Source
Probability of post-transplant mortality, per 12-week cycle	2.8%	Beta (0.6%)	Bernstein et al. (2018) [20]
Probability of PAH hospitalization, per 12 week cycle			
Low risk	1.8%	Beta (0.3%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). COMPERA
Intermediate-low risk	2.7%	Beta (0.2%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). COMPERA
Intermediate-high risk	4.3%	Beta (0.2%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). COMPERA
High risk	5.7%	Beta (0.6%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). COMPERA
Lung/heart-lung transplant	1.8%	Beta (0.4%)	Assumption[†]
HR of PAH hospitalization, sotatercept plus background therapy vs background therapy alone	0.08	Log-normal (1.05)[*]	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR
Baseline use of infused prostacyclin	39.6%	Beta (8.0%)	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR
Relative change in infused prostacyclin use			
Low risk	− 90.0%	Normal (18.0%)	Assumption[‡]
Intermediate-low risk	0.0%	Normal (0.0%)	Assumption[‡]
Intermediate-high risk	48.8%	Normal (9.8%)	Pizzicato et al. (2022) [34]
High risk	58.1%	Normal (11.6%)	Pizzicato et al. (2022) [34]
Lung/heart-lung transplant	− 100.0%	Not varied	Assumption[§]
RR for infused prostacyclin escalation	0.33	Log-normal (0.40)[*]	Merck & Co., Inc., Rahway, NJ, USA, data on file (2023). STELLAR

Table 2 continued

Parameter	Value	Distribution (SE) for probabilistic analysis	Source
Time cut-off for treatment (de-)escalation, week	52.2	Normal (10.4)	Assumption

Source: STELLAR database date: 06 DEC 2022
*Standard errors for log-transformed HRs or RRs are presented
**This assumption is in line with the 2022 treatment guideline for pulmonary hypertension [15]
†It was assumed that the probability of PAH hospitalization among those with lung/heart-lung transplant was the same as that for the low-risk group, given clinical experts' opinions that patients would likely have better prognosis after the lung/heart-lung transplant
‡This assumption was validated via consultation with clinical experts and examined in sensitivity analyses (Table 4)
§This assumption was validated via consultation with clinical experts
HR hazard ratio; *PAH* pulmonary arterial hypertension; *RR* relative risk; *SE* standard error

(Supplementary Figure 1–3), one of the largest and most representative, prospective multicenter registries in PAH.

In the base case, it was assumed that the addition of sotatercept to background therapy would have an effect on mortality. Specifically, the direct treatment effect observed in the overall STELLAR population (HR for all-cause mortality = 0.25; 95% CI 0.05–1.19 per Merck & Co., Inc., Rahway, NJ, USA, data on file 2023, including a mixed population with any of the four risk strata) was applied to the four risk health states but not the lung/heart-lung transplant state. This assumption was based on the clinical understanding that after transplantation patients would be free of PAH and then discontinue PAH therapy [18, 19].

In summary, any difference in predicted LYs between treatment arms would be driven (1) indirectly by any difference in the estimated proportion of patients staying in each non-death health state and (2) directly by the observed treatment effect in STELLAR. The robustness of observed treatment effect was examined in sensitivity analyses.

Lung/Heart-lung Transplant by Risk Stratum and Post-transplant Mortality

Patients who have reached maximal PAH pharmacotherapy may receive a lung/heart-lung transplant as an end-stage therapeutic option [15]. As per current treatment standards, such as the 2022 ESC/ERS guidelines, we assumed that only patients with intermediate-high or high risk would be eligible for lung/heart-lung transplant (Table 2) [15]. Patients transitioning to the lung/heart-lung transplant health state would remain in this state until death. COMPERA informed lung/heart-lung transplant probabilities. Post-transplant mortality was derived from the literature [20].

PAH Hospitalization by Risk Stratum

Patients with PAH often require hospitalization. We modeled PAH hospitalizations by applying risk stratum-adjusted probabilities of PAH hospitalization to each health state, with higher-risk patients at greater risk of hospitalization. These probabilities were based on observed data from the COMPERA registry (Table 2).

Infused Prostacyclin Escalation and De-escalation by Risk Stratum

Patients with PAH may experience escalation or de-escalation in PAH therapy over the course of the disease [15]. The addition of infused prostacyclin is recommended for patients with intermediate-high or high risk [15].

Table 3 Deterministic and probabilistic base case results

	Sotatercept plus background therapy	Background therapy alone	Difference	Interpretation
Deterministic outcome				
Total LYs, per patient				
By health state	16.5	5.1	11.5	Favoring sotatercept plus background therapy
Low risk	13.3	1.5	11.9	
Intermediate-low risk	2.7	1.2	1.6	
Intermediate-high risk	0.5	0.8	− 0.4	
High risk	0.04	1.6	− 1.5	
Lung/heart-lung transplant	0.01	0.04	− 0.03	
By infused prostacyclin use	16.5	5.1	11.5	Favoring sotatercept plus background therapy
Infused prostacyclin-use	1.9	2.0	− 0.1	
Infused prostacyclin-free	14.7	3.1	11.6	
Landmark survival				
1 year	96.4%	91.9%	4.5%	Favoring sotatercept plus background therapy
3 years	89.1%	67.2%	21.9%	
5 years	83.0%	45.9%	37.2%	
7 years	76.5%	27.1%	49.3%	
10 years	67.5%	11.0%	56.5%	
15 years	53.5%	1.8%	51.7%	
20 years	41.5%	0.3%	41.2%	
25 years	30.2%	0.1%	30.2%	
30 years	20.5%	0.0%	20.5%	
Clinical events, per 1000 patients				
PAH hospitalization	121	804	683**	Favoring sotatercept plus background therapy
Lung/heart-lung transplant	1	6	4**	Favoring sotatercept plus background therapy
Probabilistic outcome, mean (95% CrI)				
Total LYs, per patient	15.9 (10.3, 18.0)	5.3 (4.3, 6.8)	10.6 (4.9, 13.2)	Favoring sotatercept plus background therapy

Table 3 continued

	Sotatercept plus background therapy	Background therapy alone	Difference	Interpretation
Total infused prostacyclin–free LYs, per patient	13.9 (8.9, 16.5)	3.3 (2.2, 4.9)	10.6 (5.2, 13.6)	Favoring sotatercept plus background therapy

All outcomes were discounted at 3% annually, as commonly adopted in population health models [14]
**The number of hospitalizations and lung/heart-lung transplants averted was rounded down to the nearest integer to be conservative regarding the clinical benefit of sotatercept. Also, half-cycle correction was applied
Crl credible interval; *LY* life-year; *PAH* pulmonary arterial hypertension

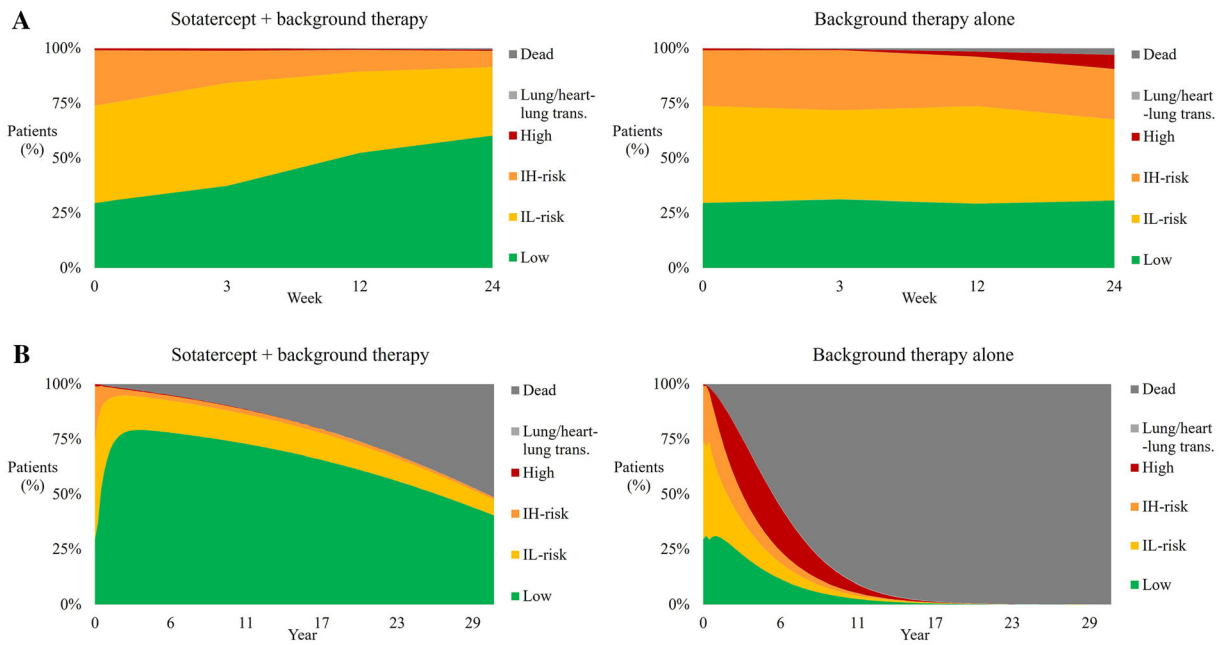

Fig. 2 Base case results, estimated patient distribution by health state. **A** Short-term distribution: Measured change in STELLAR. **B** Long-term patient distribution. The patient distribution prior to half-cycle correction is shown above. The majority (around 75% to 80%) of patients receiving sotatercept plus background therapy would experience disease improvement in the first 1–3 years and then remain in the low risk state for the remainder of their lifetime. In contrast, patients receiving background therapy alone were more likely to experience disease progression within the same time frame, with > 80% of the patients remaining in the non-low risk health state afterwards. *IH* intermediate high; *IL* intermediate low; *trans.* transplant

To reflect treatment change as the disease progresses or improves, we applied risk stratum-specific use of infused prostacyclin to each health state (Table 2). We also assumed that changes in PAH therapy would occur 1 year from model baseline, with alternative time cut-offs explored in sensitivity analyses. In the base case, it was assumed that sotatercept would have a direct and indirect effect on reducing escalation to infused prostacyclin analog use. Specifically, any predicted long-term difference in infused prostacyclin use between treatment arms would be driven (1) indirectly by any difference in the estimated proportion of patients

Table 4 Deterministic sensitivity analysis results

Parameter	Base case setting	Sensitivity analysis setting	Incremental outcomes between sotatercept plus background therapy and background therapy alone			
			Total LYs gained per patient	Infused prostacyclin–free LYs gained per patient	PAH hospitalizations averted per 1000 patients*	Lung/heart-lung transplants averted per 1000 patients*
Base case	–	–	11.5	11.6	683	4
Time horizon	30 years	3 years	0.3	0.7	342	1
		5 years	0.9	1.5	502	3
		15 years	6.2	6.8	714	5
Annual discounting for outcomes beyond the first model year	3.0%	0.0%	18.2	17.9	740	5
		5.0%	8.7	9.0	644	4
Probability of lung/heart-lung transplant, intermediate-high risk*	0.05%	0.03%	11.5	11.6	683	4
		0.07%	11.5	11.6	682	5
Probability of lung/heart-lung transplant, high risk*	0.07%	0.04%	11.5	11.6	683	3
		0.10%	11.5	11.6	683	6
HR of PAH hospitalization, sotatercept vs placebo*	0.08	0.01	11.5	11.6	788	4
		0.62	11.5	11.6	− 122	4
HR of all-cause mortality, sotatercept vs placebo*	0.25	0.05	12.9	12.9	672	4
		1.19	5.4	6.2	727	5
RR of infused prostacyclin escalation	0.33	0.15	11.5	11.7	683	4
		0.71	11.5	11.5	683	4

Table 4 continued

Parameter	Base case setting	Sensitivity analysis setting	Incremental outcomes between sotatercept plus background therapy and background therapy alone			
			Total LYs gained per patient	Infused prostacyclin–free LYs gained per patient	PAH hospitalizations averted per 1000 patients*	Lung/heart-lung transplants averted per 1000 patients*
Data source for long-term (starting week 24) risk strata transition probabilities	Observed STELLAR data within week 12 to 24 (i.e., last observation carried forward)	Observed STELLAR data within week 24 to 36	8.9	9.3	783	0
Parametric distribution for the parametric survival model for all-cause mortality	Gompertz	Weibull**	12.0	12.0	671	4
Data source for all-cause mortality inputs	COMPERA data from Rosenkranz et al. (2023), fitted using a dependent Gompertz model	COMPERA data from Boucly et al. (2022), fitted using a dependent Gompertz model**	9.8	9.7	497	3
		COMPERA data from Boucly et al. (2022), fitted using a dependent Weibull model**	11.6	11.3	485	2

Table 4 continued

Parameter	Base case setting	Sensitivity analysis setting	Incremental outcomes between sotatercept plus background therapy and background therapy alone			
			Total LYs gained per patient	Infused prostacyclin–free LYs gained per patient	PAH hospitalizations averted per 1000 patients*	Lung/heart-lung transplants averted per 1000 patients*
Data source for all-cause mortality in patients with sustained low risk	COMPERA data from Rosenkranz et al. (2023), fitted using a dependent Gompertz model throughout the lifetime horizon	General mortality applied 1 year after baseline	11.7	11.8	697	5
		General mortality applied 2 years after baseline	11.7	11.8	694	5
		General mortality applied 3 years after baseline	11.7	11.8	691	4
		General mortality applied 5 years after baseline	11.7	11.8	687	4
Time cut-off for treatment escalation and de-escalation	1 year	0.5 year	11.5	11.7	683	4
		2 years	11.5	11.4	683	4
		5 years	11.5	10.6	683	4

Table 4 continued

Parameter	Base case setting	Sensitivity analysis setting	Incremental outcomes between sotatercept plus background therapy and background therapy alone			
			Total LYs gained per patient	Infused prostacyclin–free LYs gained per patient	PAH hospitalizations averted per 1000 patients*	Lung/heart-lung transplants averted per 1000 patients*
Relative change in the proportion of patients on infused prostacyclin in patients with sustained (≥ 1 years since baseline) low risk	90% reduction	100% reduction (i.e., no patients with sustained low risk would be on any infused prostacyclin)	11.5	12.1	683	4
		80% reduction	11.5	11.2	683	4
Relative change in the proportion of patients on infused prostacyclin in patients with sustained (≥ 1 years since baseline) intermediate-high risk	48.8% increase	40% increase	11.5	11.6	683	4
		60% increase	11.5	11.6	683	4

Table 4 continued

Parameter	Base case setting	Sensitivity analysis setting	Incremental outcomes between sotatercept plus background therapy and background therapy alone			
			Total LYs gained per patient	Infused prostacyclin–free LYs gained per patient	PAH hospitalizations averted per 1000 patients*	Lung/heart-lung transplants averted per 1000 patients*
Relative change in the proportion of patients on infused prostacyclin in patients with sustained (≥ 1 years since baseline) high risk	58.1% increase	50% increase	11.5	11.6	683	4
		70% increase	11.5	12.7	683	4

*The number of hospitalizations and lung/heart-lung transplants averted was rounded down to the nearest integer to be conservative regarding the clinical benefit of sotatercept

**Varied within the 95% confidence interval

†Among all the parametric survival models (with risk strata as covariates) fitted to COMPERA survival data (from either Rosenkranz et al. or Boucly et al.), only the Gompertz and Weibull models were associated with monotonically increasing hazards and therefore deemed clinically plausible by the clinical experts we consulted. The other models (including exponential, log-normal, log-logistic, gamma, and generalized gamma models) were associated with either constant or monotonically decreasing hazards that were considered clinically implausible

HR hazard ratio; *LY* life-year; *PAH* pulmonary arterial hypertension; *RR* relative risk

staying in each risk strata health state and (2) directly by the observed treatment effect in STELLAR (risk ratio = 0.33; 95% CI 0.15–0.71 per Merck & Co., Inc., Rahway, NJ, USA, data on file 2023). The uncertainty of observed treatment effect was examined in sensitivity analyses. This analysis technique was utilized in order to increase the robustness of a model.

Model Outcomes

Primary outcomes included: (1) per patient total LYs, (2) per patient infused prostacyclin–free LYs, and (3) the number of PAH hospitalizations and lung/heart-lung transplants per 1000 patients.

Sensitivity Analyses

Deterministic sensitivity analyses (DSAs) and probabilistic sensitivity analyses (PSAs) were conducted to test the uncertainty of the deterministic base case results. DSAs were conducted by varying one parameter at a time. PSA were conducted by simultaneously drawing values for each parameter based on its point estimate and variance information (Table 2). The PSAs were run 1000 times to increase stability of results.

RESULTS

Base Case

Over a 30-year lifetime horizon, the presented model predicted a total of 16.5 years of life in patients receiving sotatercept plus background therapy compared with 5.1 years of life for patients receiving background therapy alone, yielding an additional 11.5 years of life expectancy per patient (Table 3 and Fig. 2).

The infused prostacyclin-free LYs (i.e., years of life during which patients did not have to receive infused prostacyclin) was 14.7 vs 3.1 for sotatercept plus background therapy vs background therapy alone, leading to 11.6 infused prostacyclin-free LYs gained with sotatercept (Table 3).

Sotatercept plus background therapy was also associated with a reduction of 683 PAH hospitalizations and 4 lung/heart-lung transplants per 1000 patients compared with background therapy alone (Table 3).

Sensitivity Analyses

The DSA demonstrated that variations in (1) the time horizon, (2) direct treatment effect of sotatercept on reducing mortality and (3) annual discounting beyond the first model year were the only model parameters with a meaningful impact on model outcomes. Other variations in model inputs had limited impact. Specifically, sotatercept was estimated to confer greater health benefits over a longer time horizon (e.g., 30 years compared to 5 years), while the inclusion of the direct treatment effect of sotatercept on mortality reduction resulted in substantially more favorable outcomes for sotatercept compared with the base case (Table 4).

Additionally, the PSA results were consistent with those of the deterministic base case analysis, indicating its stability (Table 3).

DISCUSSION

Our analysis suggests that adding sotatercept to background therapy is expected to increase life expectancy by approximately 12 years among patients with PAH, while reducing utilization of infused prostacyclin, PAH hospitalizations, and lung/heart-lung transplants compared to background therapy alone.

Sensitivity analyses showed that the results of the base case analysis were robust. In the DSA in which time horizon, the direct treatment effect of sotatercept on reducing mortality, and annual discounting beyond the first model year were varied, sotatercept plus background therapy was consistently associated with fewer all-cause deaths, infused prostacyclin-free life years, PAH hospitalizations, and lung/heart-lung transplants compared with background therapy alone. The PSA results showed the stability of overall results.

The longer life expectancy in sotatercept-treated patients was primarily driven by patients remaining in the low-risk health state for a longer duration.

This analysis has limitations that should be considered when interpreting the results. First, like most population health models, this study relied on trial data to predict patients' transitions across risk strata health states over a lifetime horizon. As with most PAH trials, STELLAR's short follow-up period may limit the generalizability of the risk strata transition probabilities beyond the trial period [16, 21]. Additionally, due to the limited number of observed clinical events (e.g., mortality) in STELLAR at the time of our study, long-term real-world data from COMPERA informed additional clinical inputs in our model. We also performed sensitivity analyses to examine to which extent alternative input values would affect the model results. These assessments generally suggest that these limitations may have limited impact, if any, on the model results. Therefore, the model was confirmed in its robustness. Second, our model relied heavily on STELLAR data to inform clinical inputs. Given this, the model's predictions may be more informative for PAH populations with

patient characteristics similar to those of the STELLAR trial population but may not be fully representative of all the patients with PAH, depending on factors such as geographic region, disease severity, and age at diagnosis. This limitation is common for most of the clinical population health models, especially in the context of rare diseases where increasing the study sample size to enhance generalizability presents even greater challenges. However, given that the STELLAR trial demonstrated consistent relative treatment effect of sotatercept across most prespecified subgroups [12], potential differences in patient profiles may have limited impact on the model results. Third, our model does not directly capture the impact of hospitalization on mortality, partly because of the small number of mortality events before and after hospitalization observed in STELLAR. However, given that both hospitalization and mortality rates were linked to risk strata health state occupancy, predicted survival would exhibit consistency with predicted hospitalization, resulting in clinically coherent outcomes within the model. Notably, this limitation could also lead to underestimation of the survival benefit of sotatercept, which might be explored in future modeling studies. More specifically, given that (1) sotatercept was observed to reduce the probability of PAH hospitalization in STELLAR and (2) patients who had PAH hospitalizations would have a higher mortality risk [22–24], using the observed treatment effect on reducing mortality (as currently used in our model) might not fully capture all the long-term survival benefit of sotatercept. Finally, while this study aimed at a comprehensive assessment of sotatercept's health impact, it did not account for the broader impact on patients, caregivers, or society. Examples of outcomes not included in this study include changes in health-related quality of life for patients and caregivers, costs associated with the add-on therapy to patients and society, productivity changes for patients and caregivers, and real-world treatment patterns. Future research may be warranted to assess these broader implications related to sotatercept and other emerging novel therapies, especially within the context of PAH with a substantial disease burden.

Our de novo model design exhibits notable strengths making it a highly informative tool for clinical decision-making. First, our model relies on multiple data sources, starting from STELLAR, which is distinguished from other PAH trials in its unique characterization of maximized background therapy, as well as the long-term real-world data obtained from COMPERA, one of the largest and most representative prospective multicenter PAH registries. The utilization of this extensive data set increases the external validity of our results. Second, our model relies on the most recent risk strata classification approach for assessing PAH severity and progression and thus is expected to provide more valid estimates compared to most prior models that relied on the traditional WHO FC classification to predict long-term outcomes [18, 19, 25]. More specifically, the WHO FC categories, ranging from FC I (most mild) to FC IV (most severe), have historically been widely used by clinicians to classify PAH severity, predict survival, and measure treatment effectiveness [26]. However, this clinician-rated classification approach presents certain limitations. In particular, it relies exclusively on clinicians' judgment of patient-reported symptoms, without accounting for objective measures of PAH severity, which can lead to potential interrater variation [16, 27–29]. Additionally, WHO FC employs a single variable to measure disease severity, which may not fully capture patients' prognosis [28]. As clinical assessment is essential for disease management, there was a pressing need for a more effective and objective assessment tool to provide key insights for establishing PAH prognosis and severity [16]. To address this, the 2022 ESC/ERS guidelines introduced an alternative comprehensive risk assessment strategy for the diagnosis and treatment of PH [15]. Using cut-off values, this multivariable risk assessment evaluates PAH-related prognostic indicators such as WHO FC, 6MWD results, and N-terminal pro-brain natriuretic peptide plasma levels to classify patients into four risk strata: low, intermediate-low, intermediate-high, and high risk. In contrast to the WHO FC, this risk stratification

approach emphasizes a comprehensive evaluation of multiple prognostic markers, thereby offering more objective diagnostic and prognostic information. This stratification approach has demonstrated greater sensitivity to prognostic changes in risk, as well as improved differentiation of long-term survival, as validated in multiple PAH registries [17, 30–33].

Moreover, our study's LY estimates were aligned with those generated in previously published population health models evaluating PAH treatments (Supplementary Table 3) [18, 19, 25]. Notably, these comparisons are only informative to some extent given that our model relied on a recently established risk strata endpoint, in contrast to prior models relying on WHO FC. For example, our study estimated a total of 5.05 LYs per patient, which is slightly lower than the range of estimates (5.72–7.31 LYs) from a Canadian health technology assessment for selexipag plus background therapy vs background therapy alone in PAH [18]. It was also lower than the range of estimates (7.11–9.19 LYs) in a published Australian study [25]. The lower LYs predicted in our analysis may be explained by not only difference in the modeled endpoint but also variations in the modeled population (shown in Supplementary Table 2). For example, our analysis was composed of previously treated patients with more advanced disease, resulting in a higher mortality risk compared with patients with newly diagnosed PAH in the Australian study. In addition, our study used more recent PAH mortality data from COMPERA, which is expected to result in our model producing more robust estimates.

In addition, given that our model relied heavily on STELLAR data (e.g., risk strata transition probabilities from weeks 0–3, weeks 3–12, and weeks 12–24), to ensure the robustness of our model results, we further performed an exploratory analysis using the risk strata transition probabilities over weeks 0–24 (i.e., the double-blind, placebo-controlled period) of STELLAR (Supplementary Table 4) and a model cycle length of 6 months. This exploratory analysis led to a gain of 10.7 years of life per patient, which closely aligned with the base case results of 11.5 years. This analysis further showed the stability of our model results when changing the way of incorporating STELLAR data.

To ensure the clinical plausibility of our model, the selection of parametric regression models used to inform our mortality inputs was based on goodness of fit in addition to clinical experts' opinions. Parametric regression models with monotonically increasing hazards were considered clinically plausible to inform this population health model. Further external validation was conducted by comparing our estimated LYs for the background therapy alone arm with those reported in previously published population health models and registries.

CONCLUSION

In conclusion, in this population health model, adding sotatercept to background therapy demonstrated an increased life expectancy and reduction in utilization of infused prostacyclin, PAH hospitalizations, and lung/heart-lung transplants. These findings need to be confirmed by longer-term real-world studies.

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Data Availability. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest. Rongzhe Liu, Iman Nourhussein, and David Bernotas are employees (or were employees during conduct of this study) of OPEN Health, which received funding from Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA to conduct the study. Adnan Alsumali, Janethe de Oliveira Pena, Rogier Klok, Dominik Lautsch, Jestinah Chevure and Eliana Martinez are employees of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA and hold stock in Merck & Co., Inc., Rahway, NJ, USA. Vallerie McLaughlin is a consultant for Aerovate, Altavant Sciences, Bayer, CVS Caremark, CorVista Health, Gossamer Bio, Janssen, Merck, and United Therapeutics; and has received grants from Aerovate, Altavant Sciences, Merck, Gossamer Bio, Janssen, and SoniVie; and received stocks from Clene. Christine Pausch has nothing to disclose. Marius M. Hoeper is a consultant and speaker for Acceleron Pharma, Inc, Actelion, AOP Orphan Pharmaceuticals, Bayer, Ferrer, GlaxoSmithKline, Janssen, and MSD.

Ethical Approval. This study relied on previously conducted studies and did not involve any new studies requiring the direct participation of human subjects; thus, ethics board approval was not required.

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